Momentum in the Nifty 500: What Survives in Liquid Names, and Six Enhancement Tests That Did Not Pass Their Decision Rules

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Abstract

On a snapshot-based point-in-time Nifty 500 universe, the equal-weighted 12-1 momentum decile spread (D10−D1) averages 1.74% per month. Its annualised Sharpe ratio is 0.85, with a 95% block-bootstrap interval of 0.34 to 1.50. These figures are gross of costs and cover 179 monthly rebalances from 2011 to 2026, a span that includes the period later designated as hold-out. The main test rebuilt the deciles inside the most liquid 30, 50 and 70% of names. No liquidity floor passed the registered joint screen (interval lower bound above 0.30 and decile monotonicity above 0.5), and the full universe did not pass either. Point estimates stay positive at every floor but decline as the universe narrows. Six enhancement and diagnostic hypotheses were not supported under their own decision rules, and three of them are inconclusive rather than negative. The results are historical, gross-of-cost evidence, not a validated investable strategy.

Read before interpreting results.

- **Sample.** All results use the full sample: rebalances 2011-01-31 to 2026-06-30, with returns through 2026-07-10. That includes 2022 onward, the period the project later reserved as a hold-out. The baseline ran before the hold-out rule existed. The enhancement batch and the liquidity test also report full-sample results, and M1 and M3 additionally looked once at a 2011–2021 vs. 2022–2026 split. No result here is a clean out-of-sample test.
- **Multiple testing.** In the legacy project ledger, the enhancement batch holds sub-tests 1–29: M1 1–9, M3 10–14, M4 15–18, M5 19–24, M6 25–28 and M7 29. No M2 exists in the archived sequence. The ledger reached 32 with tests outside this paper. The baseline (MOM-B) and the liquidity-universe test (MOM-L) sit outside the legacy count. Provisional Deflated Sharpe Ratios were computed at 6 and 14 trials; they are not relied on here.
- **Costs.** Every return is gross of brokerage, taxes, bid-ask spreads and market impact. The one exception is M1, which applies a simplified capital-gains tax to the long leg only. No result is net of trading costs, and no capacity analysis was done.
- **Point-in-time universe.** Membership is a snapshot-based approximation built from 33 semi-annual index snapshots, so a stock's status can be up to about six months off. The 2018-06-30 snapshot is truncated, which may misclassify up to about 22% of names in July–December 2018. Eligibility also requires a complete trailing-year return window, so the tested universe is not the whole index.
- **Other.** Seven mid-sample month-ends were dropped. Most fell on dates the panel lists although the exchange was closed. A calendar-corrected rerun is pending and could change every number in this paper. Intervals are 95% percentile block-bootstrap intervals (12-month blocks, 10,000 resamples): circular for the baseline, non-circular moving-block for the liquidity and M5 analyses. The other batch statistics are t-statistics or point estimates. Analyses added during drafting are labelled *post hoc*.

1 Introduction

Momentum is one of the best-documented cross-sectional return patterns [2, 8], and it is prone to crashes [4, 6]. This paper does not claim to discover momentum in India. It contributes a careful implementation on a snapshot-based Nifty 500 universe, organised around one practical question: *how much of the 12-1 momentum spread remains when the portfolio is built only from more liquid names?* That question matters because the baseline effect is strongest in the least-traded stocks. Two smaller contributions accompany it. The first is the baseline estimate with its uncertainty. The second is a record of six pre-registered refinements and diagnostics (M1, M3–M7) that did not pass their own criteria.

Summary of findings. The full-sample, gross baseline spread is positive, and its block-bootstrap interval excludes zero. The decile profile is broadly ordered: 7 of 9 adjacent steps rise. When deciles are rebuilt inside liquid subsets, point estimates remain positive. The Sharpe ratio falls from 0.755 (full universe, same harness) to 0.710, 0.507 and 0.394 at the top 70, 50 and 30% by turnover. No floor met the registered joint screen, so the liquidity result is suggestive, not decision-grade. None of the batch hypotheses was supported. Regime-conditional momentum is covered in Sahi [11], and path entropy (M7) in Sahi [13].

2 Data and universe

The panel is a daily feature panel: 3,328,132 rows, 4,237 dates and 1,069 tickers, from 2010-01-04 to 2026-07-10. Prices come from Yahoo Finance, with Bloomberg filling 114 tickers; the Yahoo close is split-adjusted. Construction is documented in `https://github.com/mriganksahi-beep/indian-equity-research/tree/main/data_pipeline`. Membership is a tri-state daily flag expanded from 33 semi-annual snapshots, and every test uses `flag == 1`.

Choices specific to this paper:

- **Rebalance dates.** The rebalance date is the panel's last date in each calendar month, which yields 199 month-ends. Months with fewer than 100 eligible names are dropped. That removes 12 in 2010 (warm-up) and 7 mid-sample: 2011-08-31, 2014-08-29, 2015-02-28, 2016-10-31, 2018-03-29, 2019-03-29 and 2020-11-30. Five of the seven coincide with exchange holidays on which the panel carried incomplete rows. The panel's date index contains such closed dates, so the cross-section collapses on them. 2015-02-28 was a genuine special Saturday session held for the Union Budget, with few rows in the panel. 2019-03-29 is unexplained.
- **Eligibility.** The 232 trading rows $[t-252,\, t-21]$ (inclusive) must all have valid returns. The one vendor-failure date (2025-03-18) and the row after it are excused, so the requirement becomes 232 minus the number of excused rows. Average eligible names by year range from 411 (2024) to 492.
- **Corporate actions.** Eight tickers with unadjusted split splices are removed from return paths in the batch and liquidity harnesses, but not in the Phase B baseline.
- **Liquidity.** Turnover is the mean of daily volume times close over rows $[t-21,\, t-1]$, with at least 15 valid values. Amihud illiquidity [1] is taken from a monthly file and lagged one month.

3 Method

Signal and portfolios. For stock i at rebalance t, the momentum signal is $m_{i,t} = \prod_{s=t-252}^{t-21}(1+r_{i,s}) - 1$ over trading rows. Eligible names are ranked each month; ties go to the first name in ticker order. Decile assignment is $\lceil 10\,\mathrm{rank}/N_t \rceil$. The monthly spread is $S_t = \bar{R}_{D10,t} - \bar{R}_{D1,t}$ (equal-weighted), and the Sharpe ratio is $\sqrt{12}\,\bar{S}/\hat{\sigma}_S$. Inference treats the monthly spread series

Table 1: Implementation harnesses. All are equal-weighted, use the 12-1 signal with the 232-row window, require at least 100 names, and are gross of costs.

	Phase B baseline (MOM-B)	Liquidity test (MOM-L)	M1/M3 batch harness
Dates	Rebalances 2011-01-31 to 2026-06-30	Same span	Windows ending 2011-02-28 to 2026-07-10
Months	179 (7 mid-sample skips)	179	186 (no gaps; see Section 5)
Forward return	Compounded over the stock's rows in (rebalance, next month-end]	Next 21 trading rows of that stock	Monthly mark-to-market of the held book
Missing forward return	Observed days compounded; the 9 positions with none excluded	Name excluded	Weights renormalised over names with a return
Corporate-action screen	No	Yes (8 tickers)	Yes (8 tickers)
Bootstrap	Circular block, 12 months, 10,000 draws, seed 20260724	Moving block (non-circular), 12 months, 10,000 draws, same seed	None for M1/M3 headline
Full-universe Sharpe	0.847	0.755	0.910 (M1); 0.859 (M3, volatility-matched)

as the unit. Twelve-month blocks preserve autocorrelation at up to a year, which suits a signal with a twelve-month memory.

Liquidity-universe test (MOM-L, main analysis). Each month the test keeps the top 30, 50 or 70% of eligible names by trailing turnover (primary measure) or by lowest prior-month Amihud (robustness). It then forms new momentum deciles inside that subset and computes D10−D1. The floors, the headline statistic and the joint screen were fixed before the run: Sharpe interval lower bound > 0.30 *and* Spearman monotonicity of the time-averaged decile profile > 0.5. The full curve is reported.

Enhancement batch (M1, M3–M7). Each mechanism had a registered headline statistic and decision rule, and each ran against the frozen signal. The rules are summarised in Table 4. Regression-based tests use Fama–MacBeth coefficients [7], with Newey–West standard errors at six lags [10] where the source says so.

Harnesses. The three notebooks implement the "same" strategy differently (Table 1). For that reason every comparison in this paper is made within a single harness.

Validation gates actually run.

- **Leakage.** No signal window, out of 133,281, ended on or after its rebalance; the smallest gap was 28 calendar days. Turnover windows ended before the rebalance date. The M4 test checked that its signal, gate and forward windows meet without overlapping.
- **Spot-checks.** A hand computation matched the vectorised signal exactly for GIPCL (2011-12-30), MOTHERSON (2019-06-28) and JISLJALEQS (2025-07-31).
- **PIT universe.** The flag drops RANBAXY and HDFC after their mergers (0 of 8 and 0 of 9 later months flagged in). The eligible count varies across dates (standard deviation 139.6), so the filter is live.
- **Exits.** Only 37 positions ended because the stock stopped trading (0.043% of eligible stock-months), spread evenly across deciles.

Table 2: Phase B baseline: D10−D1, equal-weighted, gross of costs, full sample of 179 rebalances. Intervals are 95% percentile intervals from a circular block bootstrap (12-month blocks, 10,000 resamples). Terciles are slices of the full-universe deciles by turnover, specific to this harness.

Statistic	Estimate	95% interval
Mean spread (% per month)	1.740	[0.769, 2.705]
Annualised Sharpe ratio ($\sqrt{12}$ convention)	0.847	[0.335, 1.503]
Annualised return / volatility (%)	19.1 / 24.7	–
Months with positive spread (%)	64.8	–
Mean monthly Spearman IC; months with IC > 0 (%)	0.0605; 66.5	[0.0350, 0.0865]
Adjacent decile steps that rise	7 of 9	–
Worst month; Jan–Sep 2020 maximum drawdown (%)	−37.3; −46.7	–
Tercile Sharpe ratios: illiquid / middle / liquid	0.88 / 0.95 / 0.42	–

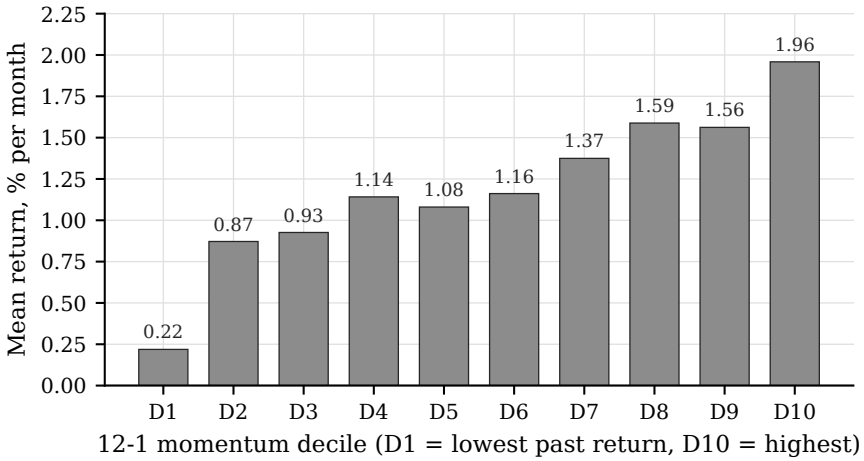


Figure 1: Mean equal-weighted monthly return by 12-1 momentum decile. Phase B baseline, full sample (179 months), gross of costs. Seven of nine adjacent steps rise.

- **Detection power.** Synthetic data with a planted signal confirmed the tests can find a real effect. M4's regression recovered a planted interaction of +0.8 as 0.801, and M6's Lind–Mehlum test found a planted hump and rejected a monotonic-concave curve.
- **Post hoc reproduction.** `make_figures.py` recomputes the decile means, the spread, the Sharpe ratio and its bootstrap interval from the archived baseline CSVs, and checks that they match the notebook output exactly.

4 Results

4.1 Baseline (MOM-B)

Table 2 and Figure 1 report the baseline. The spread interval and the IC interval both exclude zero. Decile means rise from 0.22% to 1.96% per month, but D5 is below D4 and D9 is below D8. The two end steps contribute 1.05 of the 1.74 percentage points. Figure 2 shows losing years in 2020 (the expected momentum crash, a 46.7% drawdown between January and September), 2024 and the first half of 2026.

The mutual-information study [12] reports a momentum IC of 0.085 for 2010–2021 and 0.084 for 2011–2021 in its own harness. The start date therefore does not explain the gap with the 0.0605 reported here. The gap comes from including 2022–2026, from harness differences, or both, and has not been decomposed.

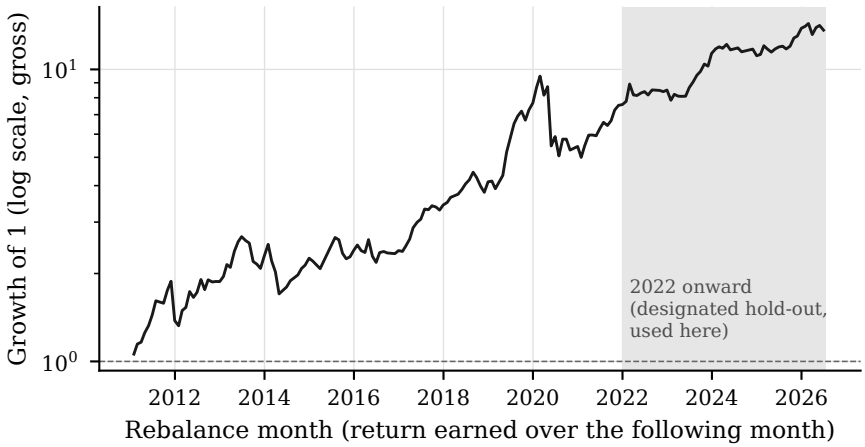


Figure 2: Cumulative D10−D1 spread on a log scale, gross of costs. The shaded area is the period the project later designated as hold-out. This baseline used it, so the figure is not out-of-sample evidence.

Table 3: Liquidity-universe test (MOM-L): deciles rebuilt within each floor, D10−D1, equal-weighted, gross, full sample, 179 months. Intervals are 95% percentile intervals from a moving-block bootstrap (12-month blocks, 10,000 resamples). Mono. is the Spearman correlation of the decile profile with decile number. The screen requires an interval lower bound above 0.30 *and* Mono. above 0.5. Ret. (post hoc) is the floor Sharpe divided by the full-universe Sharpe, a ratio of point estimates.

		Turnover filter (primary)				Amihud filter (robustness)				
Floor	Names	Sharpe	95% interval	Mono.	Ret.	Sharpe	95% interval	Mono.	Ret.	Screen
Top 30%	143	0.394	[−0.111, 0.789]	0.75	52%	0.394	[−0.089, 0.856]	0.79	52%	fail/fail
Top 50%	238	0.507	[0.016, 0.922]	0.87	67%	0.519	[−0.012, 0.991]	0.84	68%	fail/fail
Top 70%	332	0.710	[0.182, 1.157]	0.98	94%	0.622	[0.051, 1.123]	0.98	82%	fail/fail
Full	475	0.755	[0.186, 1.323]	0.94	–	0.758	[0.190, 1.326]	0.94	–	fail/fail
Mean spread, % per month. Turnover: 0.96, 1.09, 1.44, 1.54. Amihud: 0.90, 1.11, 1.25, 1.54.										

4.2 Liquidity-universe test (MOM-L)

Table 3 and Figure 3 report the main analysis. All eight filtered runs fail the registered joint screen (four floors under each of two liquidity measures). Every run, including the full universe, fails on the interval criterion, while all pass on monotonicity. Point estimates decline as the universe narrows, and the two liquidity measures agree to within 0.09 at every floor. At the top 70%, both intervals exclude zero. At the top 50%, the turnover interval only just excludes zero and the Amihud interval includes it. At the top 30%, both include zero. Intervals for different floors overlap, so the ordering across floors is a central tendency, not an established difference.

As a *post hoc* point-estimate description, the floor Sharpe ratios are 94% (turnover) and 82% (Amihud) of the same-harness full-universe value at the top 70%, and 67–68% at the top 50%. These ratios have no interval, were chosen after the results were seen, and are not evidence that liquid momentum is investable. The top-30% rebuilt universe (0.394) and the Phase B liquid tercile (0.42) are close, and they come from different harnesses. The data therefore do not support the claim that rebuilding deciles, as opposed to slicing them, changes the liquid-name result.

4.3 Enhancement batch

Table 4 summarises the batch. The notes below record what each test actually computed, and separate headline results from supporting cuts.

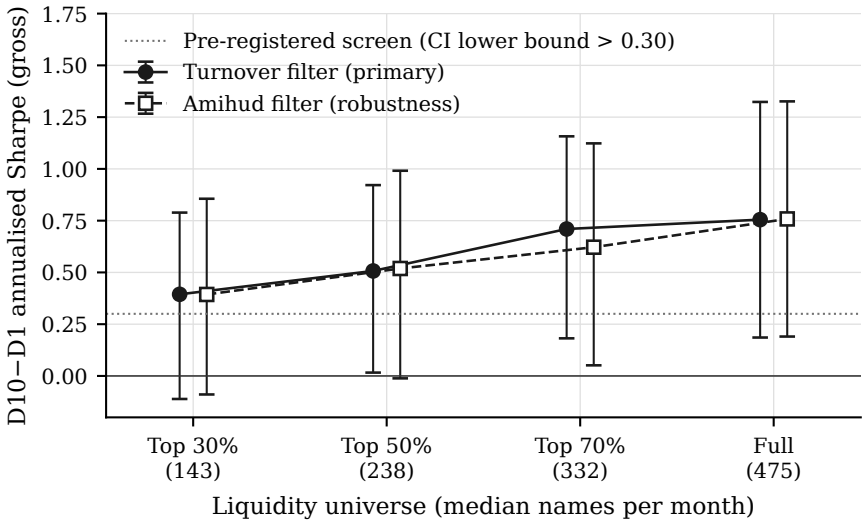


Figure 3: D10−D1 Sharpe ratio with deciles rebuilt within each liquidity universe, with 95% moving-block bootstrap intervals. Full sample, gross of costs. No floor's interval lower bound clears the registered 0.30 screen (dotted line).

M1 tracked tax lots first-in-first-out and taxed the long leg at 20% if a lot was held 365 days or less, 12.5% otherwise. Those are the rates in force since July 2024, and the model applied them across the whole sample. It had no exemption, no loss harvesting, an assumed capital of Rs 1 crore and a gross short leg, so it is not a historical tax calculation. Quarterly rebalancing kept the gross Sharpe but left 91% of exits short-term. In this model the 365-day threshold, not turnover, was the binding constraint.

M3's linear Fama–MacBeth liquidity controls left the momentum coefficient intact (104%). That reflects a specification limitation, not evidence that the effect is independent of liquidity. The explicit signal-by-illiquid interaction ($t = 1.75$) is suggestive only.

M4: the supporting cuts do not establish an effect (sign-only $t = 0.54$, 63-day horizon $t = 0.80$, no liquidity tercile above $t = 1.4$). The single form at $t = 2.04$ is one of four examined.

M5's secondary golden-cross proxy gave +0.401% per month. Its iid t-statistic is 1.68, computed as the mean over its standard error, not Newey–West. Its moving-block bootstrap interval of [+0.08, +0.71] excludes zero. The two methods disagree, so this is an exploratory secondary result, not a pass; the registered primary proxy did not pass. On crash risk, young trends had deeper maximum drawdowns than mature ones under all three proxies, including after scaling by volatility. The frequency of months below −10% was higher for young trends only under the 52-week-high proxy (8.4% vs. 1.7%). Under the 200-day moving average it was lower (2.6% vs. 3.8%), and under golden-cross age it was equal (3.2%).

M6: with finer buckets the momentum coefficient rises toward illiquid names on Amihud (0.0017 to 0.0091) and falls toward high-turnover names (0.0113 to 0.0011). The interpretation best supported by the Lind–Mehlum tests [9] is that the three-bucket "hump" arose from coarse bucketing of a roughly monotonic gradient.

M7 is reported in Sahi [13]. The permutation entropy [3] of formation-window daily returns showed no relation to forward returns ($t = -0.29$, and -0.12 after controlling for frog-in-the-pan, 5). Entropy values fell in a narrow 0.95–1.0 band, and the gate design stopped the test before any performance number was computed.

Two by-products were never registered as hypotheses: a constant-volatility overlay [4], with Sharpe 0.899 vs. 0.859, and signal-proportional weighting, with 0.898 vs. 0.767 equal-weight within the (10,8) band. Both are exploratory and would need a new, frozen test.

Table 4: Registered enhancement and diagnostic hypotheses. All are full sample and gross of transaction costs. "Reading" applies the registered rule. A result is "inconclusive" when its interval, or its t against the bar, does not exclude zero.

	Registered claim and rule	Headline evidence	Reading
M1	Rebalancing every 2, 3 or 6 months, or hysteresis bands, keeps ≥90% of gross Sharpe *and* materially cuts after-tax drag	Quarterly: 99.9% of 0.910, after-tax change +0.02% per month; semiannual: 69.8%, −0.54%; bands 78–90%, all after-tax changes negative	Not supported (point estimates; no intervals)
M3	Volatility-scaled ranking narrows the illiquid-minus-liquid tercile Sharpe gap	Gap 0.277 (plain) → 0.581 (vol-120) and 0.360 (vol-60), in M3's 186-month harness	Not supported; moved the opposite way (no interval on the gap)
M4	The skipped-month return gates momentum: Fama–MacBeth interaction, Newey–West $t > 2$	Interaction 0.00107, $t = 1.91$; 1 of 4 gate forms reached $t = 2.04$	Inconclusive under the registered threshold
M5	Younger trends outperform after removing momentum (primary proxy: 52-week high)	Young minus mature −0.021% per month, iid $t = -0.05$, moving-block 95% interval [−0.64, +0.65]	Inconclusive
M6	The mid-liquidity Sharpe peak is a genuine interior maximum (Lind–Mehlum)	0 of 4 curves (Amihud, turnover × 10 and 20 buckets) pass	Not supported
M7	Smooth (low-entropy) paths outperform at equal momentum (distinctness gate)	Entropy coefficient Newey–West $t = -0.29$; later gates not reached	Inconclusive; see Sahi [13]

5 Robustness and diagnostics

179 vs. 186 months (post hoc reconciliation). The M1 notebook attributes its seven extra months to a forward-return convention that reached later dates. The archived counts do not support that explanation. M1 splits its 186 months into 131 that end before 2022 and 55 that end on or after it. Those are exactly the number of calendar months from 2011-02 to 2021-12 and from 2022-01 to 2026-07, so M1 has no gaps and shares Phase B's first and last windows. Phase B has 124 and 55 under the same split. All seven extra M1 observations are therefore the seven interior months Phase B skipped. M1's code keeps the previous month's book and marks it to market through them (`reconcile_months.py`).

Liquidity measures. Turnover and Amihud give the same ordering. In M6 the two measures slope in opposite directions by construction while reaching the same economic conclusion.

2018 window and coverage. The mean spread over July–December 2018 was 0.462% per month ($n = 6$), against 1.784% in the other months (rough $z = 0.45$). Dropping that window did not change M6. Of all (ticker, rebalance) pairs, 62.65% have a valid signal. A looser 186-of-232-row rule would add 3.66 percentage points. In 2024 the strict window removed 18.3% of names that had a full window, against 3.9–17.9% in other years. The filter responsible has not been audited.

6 Limitations and what this paper does not establish

- **No untouched hold-out.** A test on data after July 2026, under a frozen protocol, would be needed.
- **Calendar defect.** Seven months were lost to closed or thin dates, and a closed date inside a window shifts row-counted windows by a day. The calendar-fix rerun is pending.

- **Harness differences.** No common-harness baseline exists. The cross-harness Sharpe range of 0.755–0.910 is an implementation effect of about the same size as some of the comparisons of interest.
- **Investability.** There is no cost model, no capacity analysis, and no per-floor turnover, holding count or concentration. Delisting proceeds are not modelled: Phase B compounds a stopped stock's observed days, and the liquidity harness drops names with no forward rows. Equal weighting tilts toward the smallest names in each universe.
- **Precision.** Even the full-universe Sharpe interval spans 0.19–1.32.

What this paper does not establish. It does not show that liquid-name momentum is investable or profitable after costs. It does not show that any liquidity floor beats any other. It does not show that the six refinements have zero effect: three are inconclusive. It does not show that the exploratory by-products work. And it offers no out-of-sample validation of any result.

7 Conclusion

Across the historical sample, the equal-weighted 12-1 momentum decile spread is positive, and its block-bootstrap interval excludes zero. The decile profile is broadly ordered rather than strictly monotonic. When deciles are rebuilt inside more liquid subsets, the point estimates remain positive. Uncertainty is wide, though, and none of the pre-specified liquidity floors passed the joint screen. The results include the later hold-out period, rely on different implementation harnesses and are gross of trading costs, so they are historical evidence rather than a validated investable strategy. The registered enhancement hypotheses were not supported under their stated decision rules, and the exploratory variants would need a new, frozen test.

Table 5: Reproducibility. Colab software versions were not recorded. The post hoc reproduction used Python 3.13, NumPy 2.5.3, pandas 3.0.5 and Matplotlib 3.11.2.

Item	Value
Panel	`feature_panel_with_pit_v2_2026-07-24.parquet` (3,328,132 rows); Amihud `amihud_monthly_v2_2026-07-24.parquet`
MOM-B	Run 2026-07-27; rebalances 2011-01-31 to 2026-06-30; $n = 179$; circular block bootstrap, 12 months, 10,000 draws, seed 20260724
MOM-L	Run 2026-08-05; $n = 179$ per floor; moving-block bootstrap, 12 months, 10,000 draws, seed 20260724
M1 / M3	Run 2026-07-30 / 2026-07-30–31; 186 monthly marks; single-look split 131 / 55 months
M4–M7	Run 2026-07-31 to 2026-08-04; 179 months (M4), 198 rebalances (M6); M5 bootstrap as for MOM-L
Scripts	`make_figures.py` (figures and reproduction checks); `reconcile_months.py`

Use of AI tools. Code for this project was written with the assistance of an AI coding assistant (Anthropic's Claude), working to specifications set by the author, and the manuscript was drafted with the same assistance. The author designed the research questions and decision rules and ran the main analyses. Some diagnostics added during revision were computed by the assistant from archived run outputs; these are labelled as revision diagnostics in the text. The author reviewed all code and text and takes full responsibility for the content.

Data and code availability. Code: https://github.com/mriganksahi-beep/indian-equity-research/tree/main/p1_momentum. The point-in-time panel construction is documented in https://github.com/mriganksahi-beep/indian-equity-research/tree/main/data_pipeline. Price data from Yahoo Finance and index-membership snapshots from a licensed vendor are not redistributed; the repository contains the scripts to rebuild the panel.

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